ENVS 193DS Final

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GitHub Repository: https://github.com/kierramiller/ENVS-193DS_spring-2026_final

```r
# Set up: load packages and datasets

library(tidyverse)
library(janitor)
library(readxl)
library(DHARMa)
library(AICcmodavg)
library(ggeffects)
library(readr)

nests <- read_csv("nest_data_final.csv")

personal_data <- read_csv("datasheet.csv") %>%
  clean_names()

personal_data <- personal_data %>%
  mutate(
    mg_of_caffeine = parse_number(mg_of_caffeine)
  )
```

Problem 1.

a. Transparent statistical methods

In part 1, my co-worker used a logistic regression to evaluate whether distance to the water's edge predicted the probability of American Avocet microhabitat use. In part 2, my co-worker used a chi-square test of independence to evaluate whether habitat use differed among bird species.

b. Suggestions for rewriting

Part 1: American Avocets were more likely to use some areas than others based on their proximity to the water's edge, suggesting that distance from water may influence microhabitat selection.

Part 2: American Avocets, Forster's Terns, and Black-necked stilts used habitat types differently, indicating that each species may have distinct habitat preferences within the San Francisco Bay wetland system.

c. Further analysis

One additional variable that would be beneficial to measure and that could influence the probability of American Avocet microhabitat use is water depth, this could be measured in (cm) at each observation location using a depth gauge or meter stick. By collecting data on water depth you can determine if water depth may influence habitat use because American Avocets typically forage in shallow water, the deeper depths may be too challenging or too scarce on feeding opportunities.

Problem 2. Data Analysis

a. Clean and wrangle

```
# Create cleaned dataset containing only variables needed
nests_clean <- nests %>%

  # Keep only columns needed for analysis
  select(
    case_control,
    height_cm,
    ant_species
  ) %>%

  # Keep only the three Crematogaster ant species
  filter(
    ant_species %in% c(
      "AB",
      "RRB",
      "BBR"
    )
  ) %>%

  # Reorder factor levels as instructed
  mutate(
    ant_species = factor(
      ant_species,
      levels = c(
        "AB",
        "RRB",
        "BBR"
```

```
      )
    ),

    # Add full scientific names
    ant_species_full = case_when(
      ant_species == "AB" ~ "Crematogaster sjostedti",
      ant_species == "RRB" ~ "Crematogaster mimosae",
      ant_species == "BBR" ~ "Crematogaster nigriceps"
    )
  )

# Display structure
str(nests_clean)
```

```
tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ case_control    : num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ height_cm       : num [1:270] 392 310 513 154 324 ...
 $ ant_species     : Factor w/ 3 levels "AB","RRB","BBR": 2 2 2 2 1 2 3 2 1 3 ...
 $ ant_species_full: chr [1:270] "Crematogaster mimosae" "Crematogaster mimosae" "Crematogas
```

```
# Display 10 random observations
slice_sample(
  nests_clean,
  n = 10
)
```

```
# A tibble: 10 x 4
   case_control height_cm ant_species ant_species_full
          <dbl>     <dbl> <fct>       <chr>
 1            0      109  RRB         Crematogaster mimosae
 2            0      142  RRB         Crematogaster mimosae
 3            0      183  RRB         Crematogaster mimosae
 4            1      180. RRB         Crematogaster mimosae
 5            0      228  RRB         Crematogaster mimosae
 6            0      368  RRB         Crematogaster mimosae
 7            0      114. RRB         Crematogaster mimosae
 8            0      169  RRB         Crematogaster mimosae
 9            0      116. BBR         Crematogaster nigriceps
10            0      244. RRB         Crematogaster mimosae
```

b. Response variable

The response variable is `case_control`. A value of 1 indicates that a bird nest was present in a tree, whereas a value of 0 indicates that the tree was selected as an available comparison tree without a nest. Biologically, this variable represents whether a tree was occupied as a nesting site by birds.

c. Purpose of study

Tree height is a continuous variable measured in centimeters. In Lujan et al. (2023), tree height is used as a key structural feature of acacia trees that may influence bird nest-site selection. In the Introduction (early paragraphs describing nest-site selection and habitat structure), the authors explain that birds choose nesting sites based on tree characteristics that affect predator avoidance and suitability for nesting. Taller trees may increase the probability of nest occurrence because they provide greater height above ground, improved visibility, and reduced accessibility to some predators. Tree height is operationalized in the Methods section under nest-site measurements, where tree structural traits, including height (cm), were recorded for each sampled tree.

Ant species is a categorical variable with three levels: Crematogaster sjostedti, Crematogaster mimosae, and Crematogaster nigriceps. In the Introduction (sections describing ant–acacia mutualisms and ecological interactions with birds, typically mid-Introduction paragraphs), the authors explain that different ant species vary in their defensive behavior and intensity of protection provided to host trees. These differences can influence bird nesting decisions by altering the balance between protection from predators and potential disturbance from ants. More aggressive or effective defending ant species may increase the likelihood of nest occurrence by providing safer nesting environments. Ant species identity is defined in the Methods section, where trees were classified based on the resident ant species occupying each acacia tree during field sampling.

d. Table of models

Model Number	Response Variable	Predictors	Interaction?
1	Nest occurrence	Tree height + ant species	No
2	Nest occurrence	Tree height + ant species	Yes (tree height × ant species)

e. Run the models

```r
# Model 1: additive model
m1 <- glm(
  case_control ~ height_cm + ant_species,
  data = nests_clean,
  family = binomial
)

# Model 2: interaction model
m2 <- glm(
  case_control ~ height_cm * ant_species,
  data = nests_clean,
  family = binomial
)
```

f. Select the best model

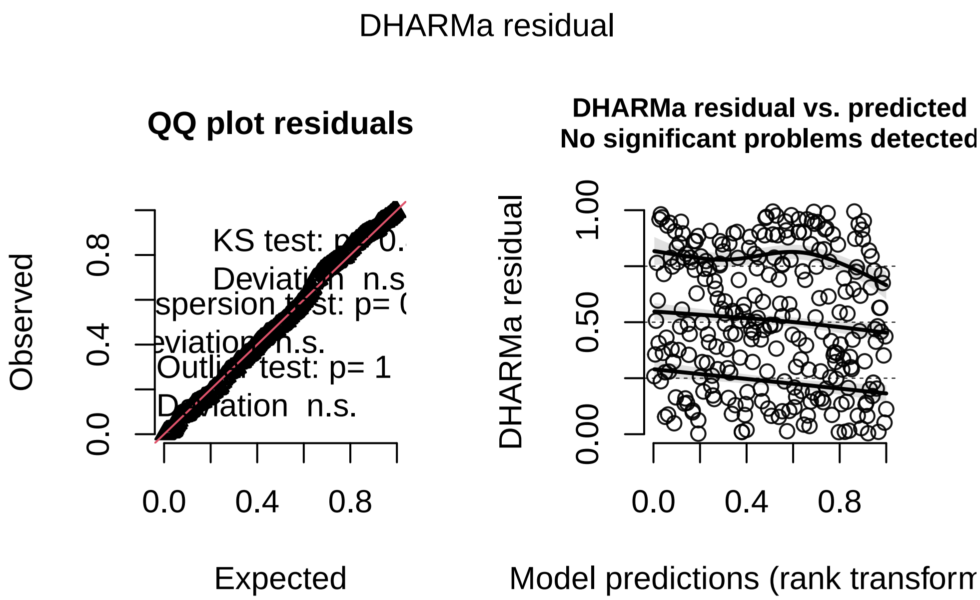
```r
# Compare AIC values to select which model is best
AIC(m1, m2)
```

```
   df      AIC
m1  4 258.7430
m2  6 237.8042
```

The best model as determined by Akaike's Information Criterion (AIC) included tree height, ant species, and the interaction between tree height and ant species as predictors of nest occurrence. This model had a lower AIC value (237.8042) than the model containing only the main effects of tree height and ant species (258.7430).

g. Check the diagnostics

```r
# Simulate residuals
sim_res <- simulateResiduals(m2)

# Diagnostic plots
plot(sim_res)
```



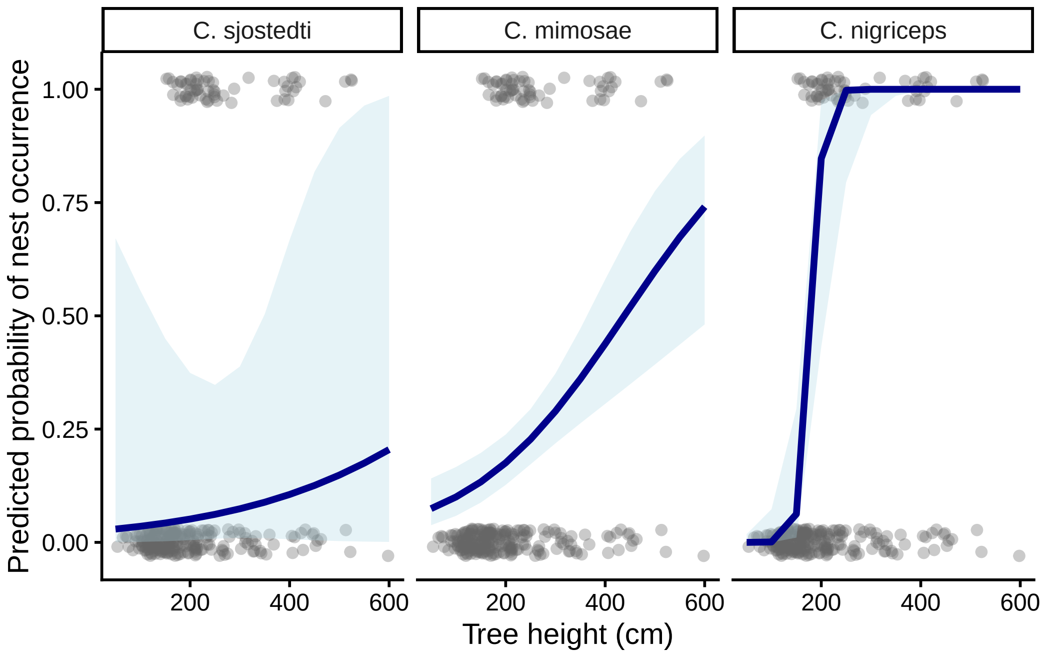
Diagnostic plots generated using simulated residuals from the DHARMa package indicated that the selected logistic regression model adequately fit the data. The residuals closely followed the expected distribution, and there was no evidence of overdispersion or influential outliers. Overall, no major violations of model assumptions were detected.

h. Visualize the model predictions

```
# Generate predicted probabilities from selected model
pred <- ggpredict(
  m2,
  terms = c(
    "height_cm",
    "ant_species"
  )
)

# Create figure showing model predictions and observed data
ggplot() +
  # Observed data
  geom_jitter(
    data = nests_clean,
    aes(
```

```
    x = height_cm,
    y = case_control
  ),
  width = 0,
  height = 0.03,
  alpha = 0.35,
  color = "gray40"
) +
# Confidence interval
geom_ribbon(
  data = pred,
  aes(
    x = x,
    ymin = conf.low,
    ymax = conf.high
  ),
  fill = "lightblue",
  alpha = 0.3
) +
# Prediction line
geom_line(
  data = pred,
  aes(
    x = x,
    y = predicted
  ),
  color = "darkblue",
  linewidth = 1.2
) +
# Separate panel for each ant species, with full scientific names
facet_wrap(
  ~group,
  labeller = labeller(
    group = c(
      "AB"  = "C. sjostedti",
      "RRB" = "C. mimosae",
      "BBR" = "C. nigriceps"
    )
  )
) +
labs(
  x = "Tree height (cm)",
```

```
    y = "Predicted probability of nest occurrence"
  ) +
  theme_classic() +
  theme(
    legend.position = "none"
  )
```



i. Figure caption

Figure 1. Predicted probability of bird nest occurrence as a function of tree height across three acacia ant species. Points represent observed nest occurrence data (0 = no nest, 1 = nest present). Solid lines represent predicted probabilities from the selected logistic regression model, and shaded regions represent 95% confidence intervals. Separate panels display model predictions for each ant species. Data from Lujan et al. (2023).

j. Calculate model predictions

```
## j. Calculate model predictions

# Predict at 600 cm first
```

```r
pred_600 <- predict(
  m2,
  newdata = data.frame(
    height_cm = 600,
    ant_species = factor(
      c("AB", "RRB", "BBR"),
      levels = levels(nests_clean$ant_species)
    )
  ),
  type = "response"
)

# Now build table AFTER prediction exists
predictions_600 <- data.frame(
  ant_species = c("AB", "RRB", "BBR"),
  predicted_probability = pred_600
)

predictions_600
```

```
  ant_species predicted_probability
1          AB             0.2048017
2         RRB             0.7407857
3         BBR             1.0000000
```

k. Interpret your results

The probability of nest occurrence differed among acacia ant species and changed with tree height (Figure 1). At the tallest tree height in the dataset (600 cm), the predicted probability of nest occurrence was approximately 0.21 for trees occupied by *Crematogaster sjostedti*, 0.74 for trees occupied by *Crematogaster mimosae*, and 1.00 for trees occupied by *Crematogaster nigriceps*. Figure 1 shows that the relationship between tree height and nest occurrence differed among ant species, which is consistent with the interaction model being selected as the best-supported model by AIC. These patterns suggest that both tree height and ant species may influence nesting decisions by birds. The differences among ant species may occur because ant species vary in their ecological interactions with birds and behaviors, making some trees safer or more suitable nesting locations than others.

Problem 3. Affective and exploratory visualizations

a. Comparing visualizations

The exploratory visualizations from Homework 2 were designed to identify patterns in the data using boxplots and scatterplots. The affective visualization from Homework 3 focused on communicating the experience and meaning behind the data through more expressive visual design choices.

All of the visualizations used the same caffeine consumption dataset and explored how caffeine intake varied with academic factors. Each visualization aimed to communicate patterns and relationships within the data.

Both the exploratory and affective visualizations suggested that caffeine consumption varied across different academic situations and tended to increase with greater workload. The patterns were similar because they were based on the same dataset, but the affective visualization emphasized the emotional impact of those patterns.

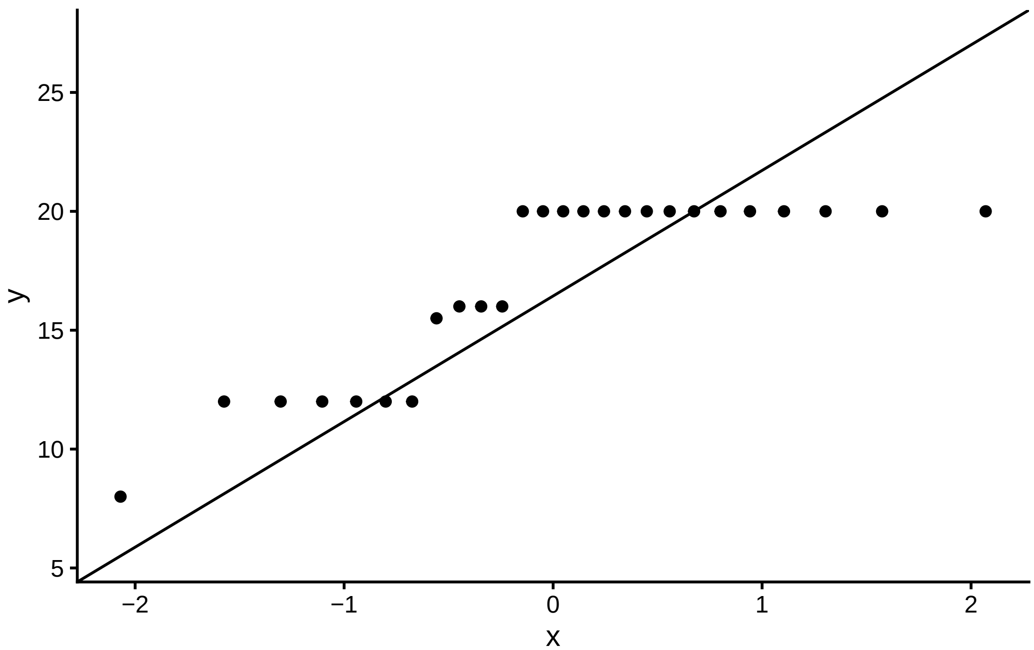
I received feedback to improve readability and make the visual message clearer through color and design choices. I incorporated this feedback by refining the layout and using visual elements that better emphasized differences in caffeine consumption compared to class location and class type.

b. Designing an analysis

Proposed Analysis: A one-way ANOVA would be an appropriate analysis because the response variable, caffeine consumption (ounces consumed), is continuous, while the predictor variable, class location, is categorical with multiple levels. This analysis would allow me to test whether mean caffeine consumption differs among the various class locations represented in my dataset.

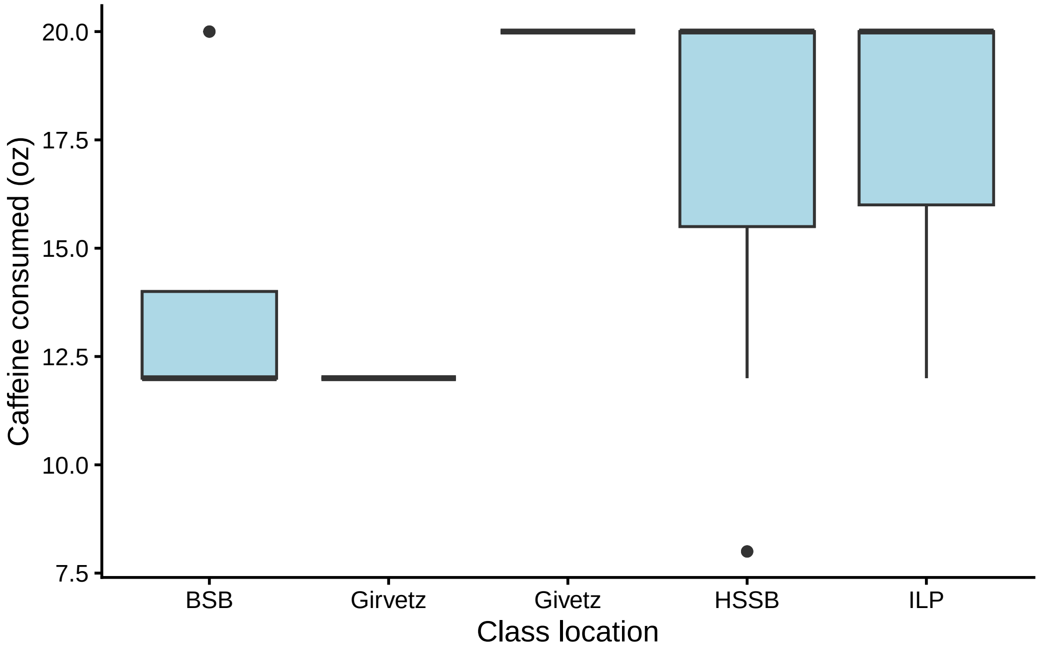
c. Check any assumptions and run your analysis

```
# Normality
ggplot(personal_data, aes(sample = ounces_consumed)) +
  stat_qq() +
  stat_qq_line() +
  theme_classic()
```



The QQ plot was used to assess whether caffeine consumption was approximately normally distributed. The points generally followed the reference line, suggesting that the normality assumption was reasonably met.

```r
# Homogeneity
ggplot(personal_data, aes(x = class_location, y = ounces_consumed)) +
  geom_boxplot(fill = "lightblue") +
  theme_classic() +

  geom_boxplot(fill = "lightblue") +
  labs(
    x = "Class location",
    y = "Caffeine consumed (oz)"
  ) +
  theme_classic()
```



Homogeneity of variance was assessed visually using boxplots. The spread of caffeine consumption appeared reasonably similar among class locations, suggesting no major violation of the equal variance assumption.

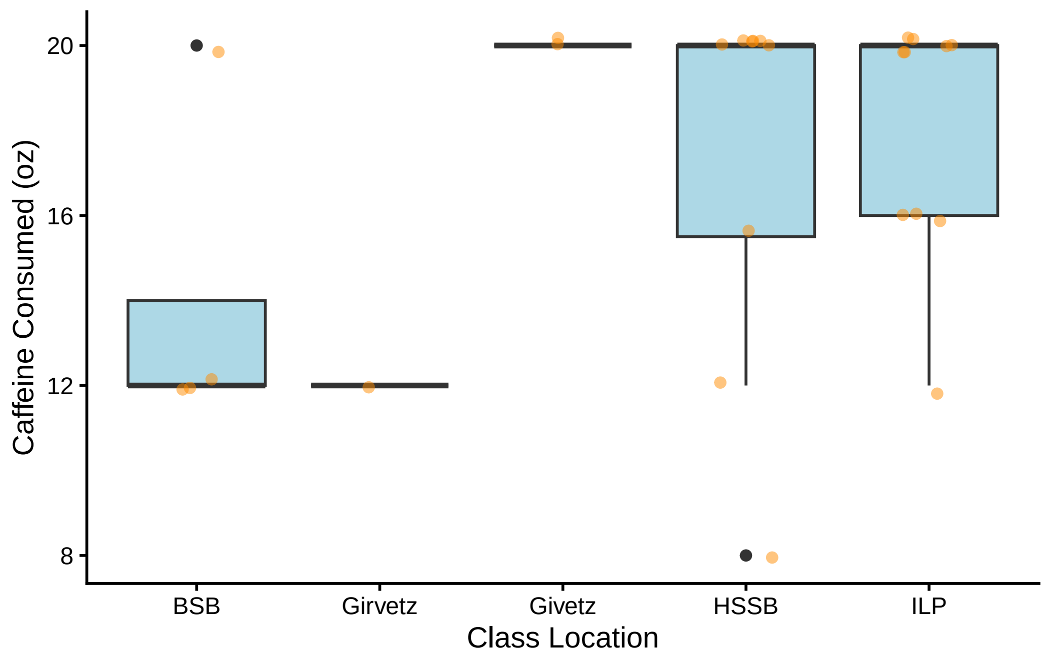
```
# One-way ANOVA

anova_model <- aov(
  ounces_consumed ~ class_location,
  data = personal_data
)

summary(anova_model)
```

```
               Df Sum Sq Mean Sq F value Pr(>F)
class_location  4  89.61   22.40   1.671  0.194
Residuals      21 281.56   13.41
```

d. Create a visualization

```
ggplot(personal_data, aes(x = class_location, y = ounces_consumed)) +
  geom_boxplot(fill = "lightblue") +
```

```
  geom_jitter(width = 0.15, alpha = 0.5, color = "darkorange") +
  labs(
    x = "Class Location",
    y = "Caffeine Consumed (oz)"
  ) +
  theme_classic()
```



e. Write a caption

Figure 2. Caffeine consumption across class locations. Individual points represent daily caffeine intake (ounces consumed), jittered to reduce overlap. Boxplots summarize the distribution within each class location group, including medians and interquartile ranges. Colors distinguish class locations, and variation in spread and central tendency suggests differences in caffeine consumption across academic environments.

f. Write about your results

Mean caffeine consumption did not differ significantly among class locations (one-way ANOVA: $F(4, 21) = 1.67$, $p = 0.194$). Although some variation in caffeine intake was observed across locations, the differences in group means were relatively small compared to within-group variation. This suggests that class location alone may not be a strong predictor of caffeine consumption, and that other factors such as time of day, workload, or number of classes may better explain differences in caffeine intake.